

From Java Foundations to Production-Grade System Architecture

This guide provides a structured transition from core Java language skills to system design and production engineering. It connects Spring Boot, Microservices, Cloud, Security, Observability, Performance, Data Systems, CAP theorem, and AI integration into a unified architecture mindset.

1. Spring Boot Internals

- 1 Dependency Injection and Bean lifecycle
- 2 Spring MVC vs WebFlux (blocking vs non-blocking)
- 3 Transaction management and AOP
- 4 Production: layered architecture (controller → service → repository)

Client → Controller → Service → Repository → DB
 ↓
 AOP / Transactions

2. Microservices Architecture

- 1 Service decomposition and bounded context
- 2 Inter-service communication (REST, messaging)
- 3 Patterns: API Gateway, Circuit Breaker, Saga
- 4 Failure handling and retries

Client → API Gateway → Microservices → DB
 ↓
 Service Mesh / Messaging

3. Cloud-Native Systems

- 1 Containers (Docker) and orchestration (Kubernetes)
- 2 Auto-scaling and load balancing
- 3 Stateless service design
- 4 AWS services: EC2, S3, RDS, Lambda

User → Load Balancer → Kubernetes Cluster → Pods → Services

4. Security

- 1 Authentication (OAuth2, JWT)
- 2 Authorization (RBAC)
- 3 API security (rate limiting, validation)

4 Spring Security filters

Client → Auth Service → Token (JWT) → API Access

5. Observability

- 1 Logging (structured logs)
- 2 Monitoring (Prometheus, Grafana)
- 3 Tracing (Jaeger)
- 4 Alerting systems

App → Logs / Metrics → Monitoring → Dashboard / Alerts

6. Performance & Scalability

- 1 Thread pools and async processing
- 2 GC tuning and memory optimization
- 3 Caching strategies
- 4 Latency vs throughput trade-offs

7. Data Systems

- 1 Caching (Redis)
- 2 Event streaming (Kafka)
- 3 Search (Elasticsearch/OpenSearch)
- 4 CQRS and Event Sourcing

8. CAP Theorem

- 1 Consistency vs Availability trade-off
- 2 CP vs AP systems
- 3 Partition tolerance as mandatory
- 4 Real-world decisions

Partition → Choose:
Consistency (CP)
OR
Availability (AP)

9. AI Integration

- 1 RAG pipelines
- 2 Vector databases

- 3 AI microservices
- 4 Scalable inference architecture

Unified Architecture Flow

